

MD. NAZRUL HUDA SHANTO

 nazrulhuda.com  +1 405-564-6370  mshanto@okstate.edu

 [Google Scholar](#)  [GitHub](#)  [LinkedIn](#)

RESEARCH SUMMARY

Researcher with 4+ years of experience across AI, systems security, and human-centered computing: an LLM agent that runs zero-knowledge proof systems in plain language, an ML system that defends cloud services from DDoS attacks automatically, and fieldwork on the digital divide in remote Indigenous communities.

Interests: Trustworthy LLM agents, security and privacy of cloud and AI systems, verifiable computation, and human-centered computing.

EDUCATION

M.S. in Computer Science, Oklahoma State University, Stillwater, OK – **GPA: 3.82/4.0** *Jan 2024 – May 2026*

B.S. in Computer Science, BRAC University, Dhaka, Bangladesh – CGPA: 3.36/4.0 *May 2018 – May 2022*

RESEARCH EXPERIENCE

Graduate Research Assistant

Jan 2024 – May 2026

Oklahoma State University

Stillwater, Oklahoma

Natural-Language Zero-Knowledge Verification via Skeptical MCP Tools [\[Code\]](#)

Distributed Intelligent Systems Lab · Advisor: Dr. Paritosh Ramanan · First-author paper under review, IEEE BigData 2026

- Built an LLM agent that lets non-experts request, check, download, and verify zero-knowledge proofs in plain language, running the CoSMeTIC zkSNARK framework (KS, LRT, and logistic-regression tests) through **8 custom MCP tools** (LangGraph, FastMCP, Flask, Redis, PostgreSQL, Docker).
- Contributed a merged pull request to the open-source CoSMeTIC zkSNARK framework: added a proof-lookup endpoint to its three prover APIs and a new input-files service, fixed a bug that made KS and LRT verification always report failure even for valid proofs, and made its Docker stack run out of the box.
- Designed the *skeptical tools* pattern: **20 server-side design decisions** that check, complete, or override the parameters an LLM sends, so the system stays correct even when the model leaves out or invents values (e.g., an interceptor replaces any model-invented user ID with the real session identity); also showed that dataset hashes and account IDs reach the model provider in every configuration, even though patient records stay inside the provers.
- Evaluated the system with a 5-configuration ablation across 3 open-weight model families (**6,000+ query runs**): task completion rose from 70.2% to **93.0%** on Qwen 3 32B, with a gain of about 23 points on every model family.
- Found that a “plausible default” inside a tool silently sent **198 status checks** to the wrong prover, so models often told users their job did not exist; replacing it with a server-side override removed all 198 errors.
- Found that our original evaluation grader wrongly marked honest error recovery as fabrication and built a corrected grader (150 of 151 agreement with manual labels); released the code, test suite, graders, and all 6,371 result records.

ML-Based DDoS Detection and Automatic Mitigation for Service Meshes [\[Prototype Code\]](#)

Department of Computer Science · Advisor: Dr. Sharmin Jahan · Paper under review, ICOIN 2027

- Designed and built a DDoS detection and mitigation framework for Kubernetes and Istio that runs as a single monitoring pod, needs no changes to application code, and collects per-pod telemetry from Envoy sidecars.
- Trained per-service classifiers (Random Forest, Gradient Boosting, SVM, Decision Tree) on **4,705** labeled traffic windows, reaching **94.0–97.8%** accuracy, with an Isolation Forest fallback for services that have no trained model.
- Built a traffic rerouting engine that patches Istio DestinationRule and VirtualService resources through the Kubernetes API to move traffic to backup versions during an attack, then restores the original routing after a cooldown.
- Ran per-pod inference in parallel, cutting detection-cycle time by **55.9%** (37.4 ms vs. 84.8 ms) against a sequential centralized baseline on a 12-pod testbed.

Undergraduate Research Assistant

Oct 2020 – Jan 2023

Computing for Sustainability and Social Good (C2SG) Lab, BRAC University

Dhaka, Bangladesh

Supervisor: Dr. Jannatun Noor

- Corresponding author on an IEEE TCC paper: designed an image compression module (27% size reduction) and built the cloud storage and retrieval framework, achieving **54% faster** load times (validated with SSIM).
- Led primary fieldwork in remote Indigenous communities of southeastern Bangladesh for HCI4D studies (CHI 2027 and Social Sciences & Humanities Open, under review; COMPASS 2026), including SPSS-based regression analysis of primary-education barriers.
- Helped simulate a resource-sharing network of 50 Nginx VMs that demonstrated **66.7%** DDoS attack mitigation (NSysS 2022).

PUBLICATIONS

UNDER REVIEW

- **M. N. H. Shanto**, P. Ramanan, “Natural-Language Zero-Knowledge Verification via Skeptical MCP Tools,” *IEEE International Conference on Big Data (BigData 2026)*, Special Session on Privacy and Security of Big Data. [\[PDF\]](#)
- M. B. Hossain, **M. N. H. Shanto**, S. Jahan, “A Decentralized ML-Based Framework for Service-Level DDoS Detection and Mitigation in Service Mesh Infrastructures,” *41st International Conference on Information Networking (ICOIN 2027)*. [\[PDF\]](#) [\[Prototype Code\]](#)
(Designed and built the full system: Envoy telemetry collection, per-service ML detection engine, and automatic Istio traffic rerouting.)
- J. Noor, **M. N. H. Shanto**, M. G. Z. A. Husna, J. Tripura, A. B. M. A. A. Islam, “Design within the ‘Back of Beyond’: Digital Divide and Opportunities among Highly-Vulnerable Ethnic Communities in Remote Nature of Bangladesh,” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*. [\[PDF\]](#)
- J. Noor, **M. N. H. Shanto**, M. G. Z. A. Husna, J. Tripura, A. B. M. A. A. Islam, “Physical Access Before Digital Access: Fieldwork Reflections from the Little-Known Indigenous Communities in Bangladesh,” *Social Sciences & Humanities Open* (Elsevier). [\[PDF\]](#)

PEER-REVIEWED PUBLICATIONS

- J. Noor, **M. N. H. Shanto**, J. J. Mondal, M. G. Hossain, S. Chellappan, A. B. M. A. A. Islam, “Orchestrating Image Retrieval and Storage over a Cloud System,” *IEEE Transactions on Cloud Computing (TCC)*, 2022. [\[PDF\]](#)
(Corresponding author; designed the image compression module and built the cloud storage and retrieval framework. Poster at NSysS 2021.)
- J. Noor, S. Biswas, T. Azhar, S. S. Karmakar, **M. N. H. Shanto**, et al., “‘Wish We Could Study in Our Mother Tongue’: Exploring Indigenous Primary Education Barriers and Proposing Guidelines in Bangladesh,” *ACM Conference on Computing and Sustainable Societies (COMPASS 2026)*. [\[DOI\]](#)
- F. F. Khan, N. M. Hossain, **M. N. H. Shanto**, S. B. Anwar, J. Noor, “Mitigating DDoS Attacks Using a Resource Sharing Network,” *9th International Conference on Networking, Systems and Security (NSysS 2022)*. [\[PDF\]](#)
(Oral presentation.)
- N. Rashid, J. Saha, R. I. Prova, N. Tasfia, **M. N. H. Shanto**, J. Noor, “Towards Devising a Fund Management System Using Blockchain,” *Computer Science & Information Technology (CS & IT)*, vol. 12, no. 18, 2022. [\[PDF\]](#)

PROFESSIONAL EXPERIENCE

Junior Software Engineer

Feb 2023 – Dec 2023

Tirzok Private Ltd.

Dhaka, Bangladesh

- Led an ML-powered document validation system for CliQloan, a mortgage loan origination system: document classification, page trimming, and key-field extraction that ran **45×** faster than manual validation (scikit-learn, AWS Textract, Lambda, ECR, Amazon MQ, Spring Boot, PostgreSQL).
- Built RESTful APIs (Node.js, MongoDB) for a government service portal serving **80,000+** citizens; optimized MongoDB aggregation pipelines to cut query latency by 40% and designed JWT-based role management.

TEACHING & SERVICE

AI Instructor, NSF Research Experiences for Teachers (RET) Program

May 2025 – Aug 2025

Department of Computer Science, Oklahoma State University

Stillwater, Oklahoma

Supervisor: [Dr. Rittika Shamsuddin](#)

- Taught 10 K–12 teachers Python, machine learning, and Transformers, guided their research projects, and designed a Python-to-LLM curriculum for their classrooms.

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, ReAct agents, Model Context Protocol (FastMCP), PyTorch, scikit-learn, LIME
Security & Verification: Zero-knowledge proofs (zkSNARKs, Sparse Merkle Trees), JWT-based access control
Cloud & Systems: Kubernetes, Istio/Envoy, Docker, Prometheus, Redis, PostgreSQL, MongoDB, RabbitMQ, AWS
Languages & Frameworks: Python, Java, TypeScript/JavaScript, SQL, Bash; Flask, FastAPI, Spring Boot, React

REFERENCES

Dr. Paritosh Ramanan

Advisor: zero-knowledge verification and LLM agents

Assistant Professor, Oklahoma State University

paritosh.ramanan@okstate.edu

Dr. Sharmin Jahan

Advisor: DDoS detection for service meshes

Assistant Professor, Oklahoma State University

sharmin.jahan@okstate.edu

Dr. Rittika Shamsuddin

Supervisor: NSF RET Program

Assistant Professor, Oklahoma State University

r.shamsuddin@okstate.edu

Dr. Jannatun Noor

Undergraduate supervisor; Director, C2SG Research Group

Associate Professor, United International University

jannatun@cse.uiu.ac.bd